RingMamba: Remote Sensing Multisensor Pretraining With Visual State Space Model

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***Abstract*—Previous studies on remote sensing foundation models have demonstrated the representational ability of convolutional neural networks (CNNs) and vision transformers (ViTs). However, these models either focus on single-sensor data, thereby overlooking complementary information across sensors, or they increase the computational complexity in pursuit of a global receptive field. In this article, we introduce RingMamba, a novel multisensor remote sensing (RS) foundation model that excels in both performance and efficiency. To achieve unified representation learning for multisensor data, we propose a multisensor self-supervised pretraining framework that integrates generative and contrastive learning strategies based on texture feature constraints, enabling unsupervised training of massive multisensor data. Considering the multiangle and multisize distribution of instances, we also propose an asymmetric scan and scan couple (ASSC) block to achieve multidirectional scanning and recognition. We validate the performance of the model on multimodal semantic segmentation, scene classification, object detection, and change detection tasks using nine public datasets. The experimental results show leading performance across these tasks. In addition, we evaluate the computational consumption of the model for classification tasks under different input sizes. The results show that our model maintains leading performance with higher throughput and lower FLOPs, especially under high-resolution inputs, making it more suitable for the remote sensing field.**

***Index Terms*—Foundation model, Mamba, multisensor, remote sensing (RS), self-supervised learning.**

I. Introduction

With the rapid development of satellite technology, the remote sensing (RS) data generated every day in the world covers hundreds of millions of square kilometers [1], [2], [3]. From the perspective of imaging methods, it covers a variety of sensors such as optical sensors and microwave sensors, providing multidimensional spatiotemporal spectral descriptions of ground objects and rich information for numerous applications. For example, optical images are usually high-resolution and can capture the shape, color, and texture details of objects in good lighting conditions. In contrast, SAR images have all-weather and all-day observation capabilities, providing insights into shape, reflectivity, and motion characteristics even in complex meteorological conditions. Therefore, the fusion of optical and SAR image data can achieve information complementarity between different types of sensors.

Many studies have demonstrated the effectiveness of multisensor data fusion, which can provide more comprehensive information about objects [4], [5], [6], [7]. However, remote sensing encompasses a wide range of tasks, such as scene classification, object detection, semantic segmentation, and change detection. Traditional dedicated models often struggle to adapt to different application scenarios. Consequently, there is an urgent need to develop a universal model capable of performing well across multiple sensor data types and tasks, thereby enhancing the efficiency and accuracy of remote sensing data analysis.

Previous studies on remote sensing foundation models have verified their generalization performance in various tasks [1], [8], [9], [10], [11], but they can still be further optimized in two key aspects.

A. Self-Supervised Learning Strategies

Most remote sensing foundation models are typically designed for single-sensor data, utilizing self-supervised methods that focus on either reconstructing pixel values of images (generative methods) or leveraging the similarity of visual features (contrastive methods). However, data from different modalities possess unique characteristics, making it challenging to adequately represent the data using only visual features. While there are foundation models proposed for multisensor data, they often ignore the distinctive characteristics of RS data, such as texture features and structural features, during the self-supervised training process. These features are crucial for accurately learning and representing RS targets. Thus, improving the incorporation of these specific characteristics can enhance the effectiveness of foundation models for multimodal RS tasks.

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B. Balance of Performance and Computational Complexity

Large-scale convolutional neural networks (CNNs) and transformer modules have been widely used in foundation models. Although these methods have achieved promising results, accurately recognizing complex scenes with varying spatial and temporal resolutions remains challenging. This is partly owing to the restricted receptive field of the CNN architecture, which hampers the model's global modeling capabilities [12]. The vision transformer addresses the restricted receptive field problem of CNNs by utilizing the self-attention mechanism, which effectively models the relationship between any region and the entire image [13]. However, the computational complexity of transformers scales quadratically with the input sequence length, leading to high computational overhead when processing downstream dense prediction tasks like change detection.

In this article, we introduce RingMamba, a multisensor foundation model incorporating an efficient Mamba module. RingMamba is pretrained on 3 million multisensor remote sensing data, including both paired and unpaired data from high-resolution optical and SAR satellites. To improve the model's ability to represent features, we developed multisensor masked image modeling and contrastive learning for self-supervised training. Specifically, for SAR images, using pixel values as prediction targets can lead to overfitting to local statistics. To address this, we introduce texture descriptor local binary patterns (LBPs) features [14] as prediction targets, which capture the local shape and appearance of the target with low computational overhead. Considering the small size and multiangle arrangement of targets in remote sensing images, we designed the asymmetric scan and scan couple (ASSC) Mamba and incorporated a rotational scanning mechanism to aid in detecting targets from various angles. Finally, we validated the generalization performance of the model on various tasks, such as multimodal semantic segmentation, change detection, scene classification, and object detection. The innovations are summarized as follows.

1) We introduce an innovative multisensor foundation model for remote sensing, RingMamba, designed for efficient joint representation learning. This approach considers the unique characteristics of different sensor data, enabling effective cross-modal data representation.
2) The RingMamba model introduces three technical innovations: a) a unified multisensor self-supervised pretraining framework that combines generative and contrastive learning strategies; b) the use of texture features as the prediction target in the generative self-supervised method, mitigating the limitations associated with pixel values; and c) the introduction of asymmetric scan (AS) couple in the RingMamba model to enhance the recognition of multiangle targets.
3) We compared RingMamba with leading foundation models and other state-of-the-art (SOTA) methods across nine public RS datasets. As shown in Fig. 1, the proposed model not only achieved competitive performance on most of these datasets but also offers higher throughput, lower FLOPs, and fewer parameters. We hope this model will inspire further research into low-power solutions for foundational models in the RS field.

II. Related Work

A. Foundation Models in the Field of Remote Sensing

Remote sensing foundation models are essential for processing a wide range of complex remote sensing data [1], [15], [16], [17], [18], [19]. Recently, self-supervised learning techniques have been widely used to pretrain these models, enabling them to acquire robust feature representations from unlabeled data. This improves their suitability for a range of downstream tasks, such as land cover classification, change detection, and object detection. The primary foundation models in the RS domain are largely based on two fundamental technologies: generative learning and contrastive learning.

Generative learning involves modeling the data distribution and recreating the input data. Currently, generative self-supervised methods are mainly based on masked autoencoders (MAEs), which pretrain the encoder by masking the image, feeding the unmasked part into a transformer-based encoder, and then letting the decoder reconstruct the input image. For example, the work of RingMo [1] demonstrated the use of generative SSL methods for optical remote sensing data, where the high resolution allows for effective pixel-level reconstruction. RingMo-Sense [16] introduced a triple-branch networks to construct spatial, temporal, and spatiotemporal information, which achieves stable long-range prediction. Scale-MAE [19] and Cross-Scale MAE were also masked reconstruction model that learns the relationship between data of different scales during pretraining. By masking parts of the images and incorporating temporal information, SatMAE [8] independently masking patches across the temporal/spectral dimension in satellite imagery. A2-MAE [20] designed a geographic encoding module to learn spatial–temporal–spectral information. SpectralMAE [21] employed 3-D masking in spectral data to integrate local spectral continuity and spatial invariance.

Contrastive learning involves learning to distinguish between similar and dissimilar pairs of data points. In remote sensing, contrastive learning can be used to identify and highlight differences between various types of land cover or temporal changes. Skysense [22] is proposed for multisensor satellite data. It has achieved leading accuracy in multiple downstream tasks, but it has 2 billion parameters and requires relatively high data and computing resources. CROMA [23] is a framework for learning rich optical-SAR multimodal representations using contrastive self-supervision. IaI-SimCLR [24] incorporated multimodal Earth observation data into a contrastive self-supervised learning framework by forcing intermodality and intramodality similarity in the loss function. These methods can achieve self-supervised pretraining learning of remote sensing multisensor data without labeling the data.

B. Mamba-Based Models in Natural Scenes

In recent years, researchers have shown a preference for CNNs or vision transformers in computer vision domains

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[12], [25], [26]. Traditional transformer models, due to their quadratic complexity, impose significant computational demands when processing sequences. Recently, the state space model (SSM) [27] has gained prominence in artificial intelligence, demonstrating enhanced performance on long sequences with autoregressive characteristics. Building upon SSM, researchers introduced Mamba, which contrasts with transformers by exhibiting linear complexity [28].

Zhu et al. [29] were the pioneers in applying Mamba to visual tasks. To address the challenges of position-sensitive image data, they introduced vision Mamba (Vim) [30], which integrates position encoding with bidirectional scanning to effectively capture the global context of images. This approach enhances Vim's ability to understand and model spatial relationships within images, improving its performance across various visual tasks. Liu et al. [31] presented VMamba, which tackles position-sensitive issues by scanning the image space in four directions. Since then, Mamba-based methods have rapidly expanded. Various architectures tailored for processing medical images have been introduced using the Mamba module, such as Mamba-Unet [32], VM-Unet [10], U-Mamba [33], LightM-Unet [34], and SegMamba [35]. Mamba is recognized for its robust global modeling capabilities and efficient computational complexity, making it particularly suitable for various tasks.

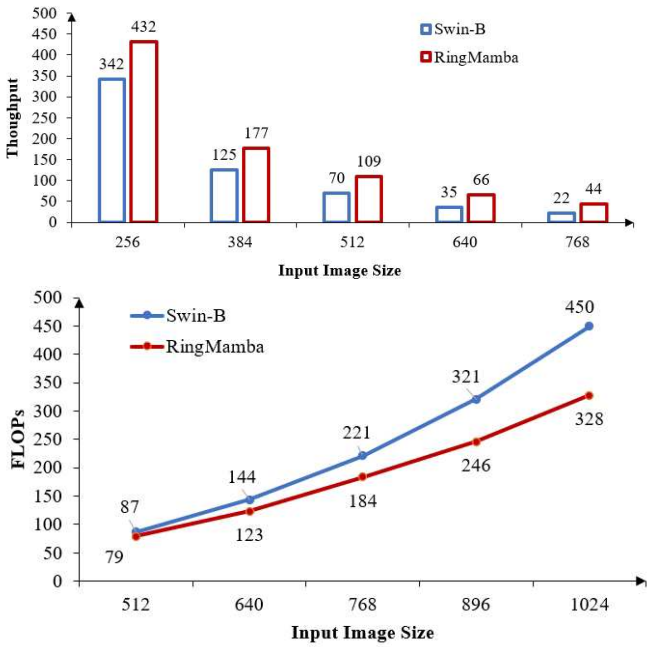


Fig. 1. Throughput and FLOPs for different image resolution inputs. These values are tested on an 80-GB NVIDIA A100 GPU, using a classification model and a batch size of 64. The window size of Swin-B increases as the input image size increases. In this figure, "RingMamba" refers to a model that incorporates LBP loss into the baseline.

C. Mamba in the Field of Remote Sensing

Recently, numerous approaches have applied Mamba to remote sensing tasks. Chen et al. [36] introduced RSMamba, which integrates shuffling with forward and backward scanning. Zhao et al. [37] implemented diagonal scanning to handle image segmentation and change detection tasks. Concurrently, Chen et al. [38] employed various scanning techniques to capture spatiotemporal relationships in bi-temporal data.

In RS image fusion, Mamba modules have been integrated into two U-shaped networks to extract and learn various spatial-spectral features. A novel module named Fusion-Mamba has been developed by extending the Mamba module to support dual inputs [39]. Xiao et al. [40] applied Mamba to facilitate multimodal fusion in a multimodal semantic segmentation task. Furthermore, leveraging Mamba's efficiency in global information modeling, a novel pan-sharpening network called Pan-Mamba has been proposed [41].

While the aforementioned methods have demonstrated promising results, most of them rely on the Mamba model to improve the accuracy of downstream tasks in the RS domain, and their generalization performance across multiple tasks remains unverified. In this article, we integrate Mamba with a multisensor self-supervised pretraining framework to create a Mamba foundation model specifically for RS data.

III. Methodology

As presented in Fig. 2, the multisensor foundation model, RingMamba, consists of two components: the multisensor pretraining framework and the ASSC Mamba. Compared to existing approaches, the proposed pretraining framework integrates multimodal masked image modeling (MM-MIM) and a multisensor contrastive learning strategy, enabling the model to effectively learn feature representations within each modality while fostering cross-modal knowledge transfer and mutual learning. To mitigate the sensitivity of pixel values in remote sensing images to variations in imaging conditions, we incorporate texture information while ensuring modal invariance, enabling more robust and generalizable feature learning. The ASSCMamba precisely extracts the semantic information from RS images. Details are shown as follows.

A. Model Architecture

1) Multimodal Fusion Encoder: We construct the input of our model as $(I_{\text{opt}}, I_{\text{sar}})$, where I_{opt} represents the optical images, and I_{sar} represents the SAR images. In each batch of input data, the ratio of paired images to unpaired images is maintained at 1:2, consistent with the ratio found in the original dataset. The overall structure of our method is shown in Fig. 2.

After remote sensing multisensor data $(I_{\text{opt}}, I_{\text{sar}})$ are input, optical and SAR data are, respectively, processed by different patch embedding features and then masked. Spatial features $(F_{\text{opt}}, F_{\text{sar}})$ can be extracted from each modality independently by deploying two mode-independent convolution modules. This design decouples spatial feature extraction from its fusion, allowing the integration of information from modal, spatial, and geographic contexts

$$\text{PE} \triangleq \text{Norm}\left(\text{Conv}_{3\times3}\left(\text{GELU}\left(\text{Norm}\left(\text{Conv}_{3\times3}\left(F\right)\right)\right)\right)\right) \quad (1)$$

$$\left\{F_{\text{opt}}, F_{\text{sar}}\right\} = \text{PE}\left(\left\{I_{\text{opt}}, I_{\text{sar}}\right\}\right) \quad (2)$$

$$\left\{F_{\text{opt}}, F_{\text{sar}}\right\} = \text{ASSCMamba}\left(\left\{F_{\text{opt}}, F_{\text{sar}}\right\} * \text{Mask}\right) \quad (3)$$

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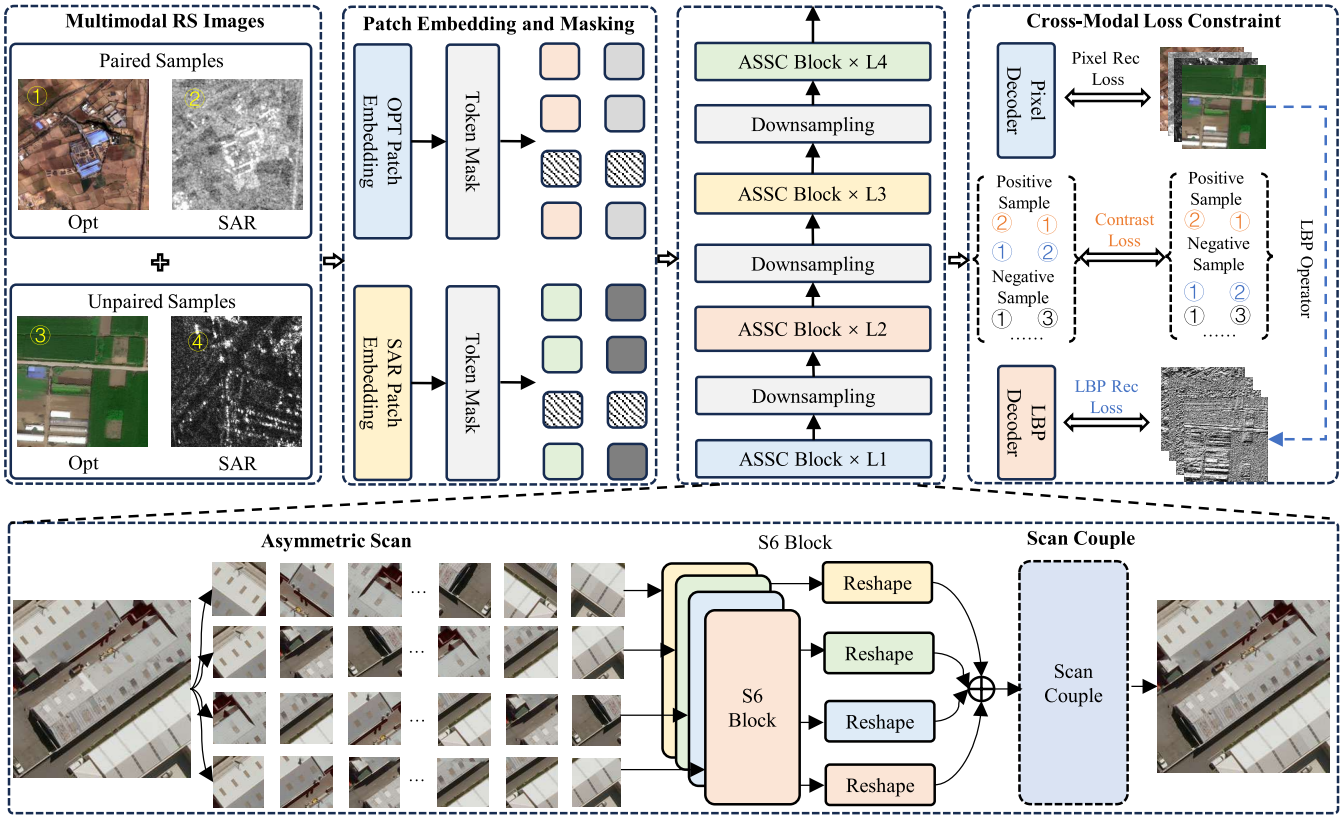


Fig. 2. Overall framework of our RingMamba. RingMamba consists of two parts: MM-MIM (top) and ASSC Mamba (bottom). MM-MIM first takes the masked multimodal images as the input. Then, it adopts two shallow convolution layers to simultaneously achieve the optical and SAR embedding. After that, it sends embedded multimodal features into ASSCMamba to achieve the projection and utilizes two shallow convolutional decoders to simultaneously reconstruct the masked pixel parts of optical and SAR images. Particularly, MM-MIM further deploys LBP and Contrast losses to better achieve the cross-modal alignment. For ASSCMamba, it mainly introduces the AS mechanism and SC module to better mine the semantic information of RS images.

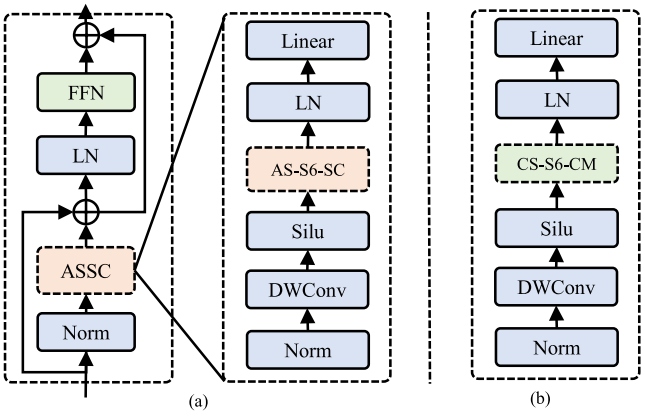


Fig. 3. Structure comparison between (a) our ASSCMamba and (b) VMamba blocks. We replace the CS and CM modules of VMamba with our AS and SC modules to better tackle RS images. S6 represents S6 block which is the core component of SSM.

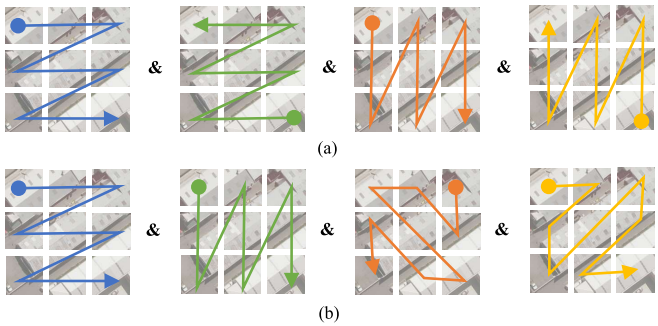


Fig. 4. Details of AS module. We introduce four asymmetric directions (horizontal, vertical, diagonal, and antidiagonal) to scan RS images, which can effectively tackle the rotation targets while keeping the similar computation costs. (a) VMamba CS. (b) Our AS.

where PE represents the patch embedding operation, and F represents the input to this process. We use two 3×3 convolutions to implement the patch embedding. ASSCMamba denotes the backbone network.

2) ASSC Block: As depicted in Figs. 2 and 3, our architecture replaces the cross scan (CS) and cross merge (CM) modules of VMamba [31] with our newly designed ASSC modules to better accommodate the unique characteristics of RS images. Although VMamba effectively handles natural scene images through its SS2D module—which includes CS, SSM (S6 block), and CM—the imaging mechanisms of RS images present distinct challenges. Specifically, ground objects in RS images tend to be smaller and more densely packed, leading to prior divergence issues with the existing CS and CM modules within the SS2D framework of VMamba. To address

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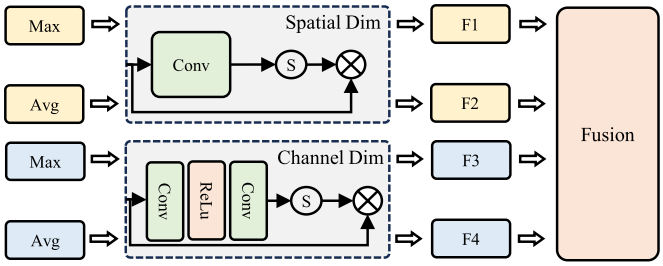


Fig. 5. Details of SC module. We achieve adaptive feature fusion by calculating the maximum and average features in both spatial and channel dimensions. This approach effectively integrates the semantic information from different scan sequences.

these challenges, as shown in Figs. 4 and 5, we introduce the AS and scan couple (SC) modules. These enhancements are designed to better capture semantic associations within RS image patch sequences, improving the ability to interpret RS data effectively.

More specifically, as depicted in Fig. 4, ground objects in RS images often have various rotation angles. The CS in VMamba only uses horizontal and vertical directions, limiting the perception of rotated targets. To address this, we introduce four AS directions—horizontal, vertical, diagonal, and antidiagonal—to scan RS images, thoroughly exploring different combinations of patch sequences while maintaining similar computational costs.

Besides, in the original CM module, VMamba directly reshapes and combines these four scanned sequences as the output. While straightforward, this approach struggles to effectively integrate the semantic information from different scan sequences. Therefore, as shown in Fig. 5, we further deploy a shallow convolution module to perform scan coupling. First, we apply average (Avg) and maximum (Max) pooling operations to extract features corresponding to scan sequences in both spatial and channel dimensions. Then, we use four shallow convolution layers and Sigmoid activation to project the attention weights, followed by skip connections and a single-layer convolution to achieve adaptive fusion. We named the backbone network with the ASSC block as the ASSCMamba.

In conclusion, the process of our proposed strategy can be described as follows:

$$\left\{F_{\text{scan}}^{i} | i = 1, 2, 3, 4\right\} = \text{AS}\left(F_{\text{img}}\right) \tag{4}$$

$$F_{\text{scan}}^{i} = \text{S6}^{i}\left(F_{\text{scan}}^{i}\right) \tag{5}$$

$$F_{\text{img}} = \text{SC}\left(F_{\text{scan}}^{1}, F_{\text{scan}}^{2}, F_{\text{scan}}^{3}, F_{\text{scan}}^{4}\right) \tag{6}$$

where F_{img} represents the input image feature, AS represents the asymmetric scan module, S6 represents the S6 block, and SC represents the scan couple module.

B. Multisensor Pretraining Framework

1) Multisensor Masked Image Modeling: Unlike existing masked image modeling methods that focus solely on single-modal pretraining [1], [8], [19], we introduce MM-MIM to enable the learning of cross-modal feature representations, effectively bridging the gap between optical and SAR images. As shown in Fig. 2, MM-MIM initially masks input images at the token level. Then, it employs two shallow convolution layers to simultaneously generate embeddings for optical and SAR images. We adopt a random mask strategy. For the input multisensor data, masks are randomly generated at the same positions for paired data, but the positions of the masks differ for unpaired data. After the MM-MIM, the embedded multimodal features are fed into our proposed architecture to achieve effective projection.

2) Generative Self-Supervised Strategy With Dual Constraints of Texture Features and Pixels: Unlike most generative self-supervised strategies that primarily focus on pixel reconstruction [1], [8], [19], our approach considers the sensitivity of pixel values in remote sensing images to variations in imaging conditions, ensuring more robust and reliable feature learning. To mitigate this issue, we incorporate texture information with modal invariance to enhance the self-supervised training process. Particularly, compared with the other texture operators, such as HOG [42] and SIFT [43], the LBPs [44] contain the relative information of adjacent image pixels, which to some extent can guide the model to reconstruct the masked regions more easily. As illustrated in Fig. 2, we introduce both pixel and LBP texture constraints, enabling the model to more effectively extract and utilize self-supervised semantic information from the data.

Specifically, in the self-supervised learning process for both optical images and SAR images, it is essential to reconstruct not only the pixel values but also to constrain the LBP features of the images. This dual reconstruction approach ensures that the model captures both the detailed pixel information and the robust structural features provided by LBP, enhancing its overall performance and reliability. We designed the decoders for pixel and LBP constraints, respectively. The feature calculation process is as follows:

$$\text{Decoder} \triangleq \text{Reshape}\left(\text{Conv}_{1\times1}\left(F\right)\right) \tag{7}$$

$$\left\{I_{\text{opt}}^{\text{rec}}, I_{\text{sar}}^{\text{rec}}\right\} = \text{Decoder}_{\text{pixel}}\left(\left\{F_{\text{opt}}, F_{\text{sar}}\right\}\right) \tag{8}$$

$$\text{Loss}_{p} = \text{L1}\left(\left\{I_{\text{opt}}^{\text{rec}}, I_{\text{sar}}^{\text{rec}}\right\}, \left\{I_{\text{opt}}, I_{\text{sar}}\right\}\right) \tag{9}$$

where Reshape represents reshaping the feature dimension, and $\text{Decoder}_{\text{pixel}}$ represents the reconstruction operation of the pixel. Due to the differing number of channels in optical and SAR images, the decoders for these modalities do not share parameters. F_{opt} represents optical feature, F_{sar} represents SAR feature, $I_{\text{opt}}^{\text{recp}}$ represents reconstructed optical image, $I_{\text{sar}}^{\text{rec}}$ represents reconstructed SAR image, and $L1$ represents $L1$ loss

$$\left\{\text{LBP}_{\text{opt}}, \text{LBP}_{\text{sar}}\right\} = \text{LBP}\left(\left\{I_{\text{opt}}, I_{\text{sar}}\right\}\right) \tag{10}$$

$$\left\{\text{LBP}_{\text{opt}}^{\text{rec}}, \text{LBP}_{\text{sar}}^{\text{rec}}\right\} = \text{Decoder}_{\text{LBP}}\left(\left\{F_{\text{opt}}, F_{\text{sar}}\right\}\right) \tag{11}$$

$$\text{Loss}_{t} = \text{L1}\left(\left\{\text{LBP}_{\text{opt}}^{\text{rec}}, \text{LBP}_{\text{sar}}^{\text{rec}}\right\}, \left\{\text{LBP}_{\text{opt}}, \text{LBP}_{\text{sar}}\right\}\right) \tag{12}$$

where LBP represents the LBP operator, and $\text{Decoder}_{\text{LBP}}$ represents the reconstruction operation of the LBP feature. LBP_{opt} represents LBP feature of optical image, LBP_{sar} represents LBP feature of SAR image, $\text{LBP}_{\text{opt}}^{\text{recp}}$ represents reconstructed LBP feature of optical image, $\text{LBP}_{\text{sar}}^{\text{rec}}$ represents reconstructed LBP feature of SAR image, and $L1$ represents $L1$ loss.

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3) Multisensor Contrastive Learning Strategy: In addition, for the multisensor paired image features, although they contain the different modal information, they essentially still represent the ground objects in the same region. Therefore, the cross-modal feature alignment can help the model better understand and reconstruct the semantic information from different sensors. Specifically, we first take the paired optical-SAR images within a batch as the positive samples and further set the other unpaired combinations within a batch as the negative samples. Then, we introduce the contrastive loss [45] to optimize the similarity distance between these positive and negative samples, which can effectively constrain our model to understand the cross-modal RS information

$$\text{Loss}_c = \text{Contrast}\left(F_{\text{opt}}, F_{\text{sar}}\right) \tag{13}$$

$$\text{Loss}_{\text{total}} = \text{Loss}_p + \text{Loss}_t * 0.5 + \text{Loss}_c \tag{14}$$

where Contrast represents contrast loss, and $\text{Loss}_{\text{total}}$ represents the total loss of pretraining stage.

IV. Experiments

In this section, a series of experiments is conducted to compare the proposed method with popular models (RS foundation models or dedicated models based on CNNs or transformers). Our evaluation covers many tasks, including multimodal semantic segmentation, scene classification, optical/SAR object detection, optical/SAR semantic segmentation, and optical change detection tasks. Through in-depth analysis of the experiments, we can further demonstrate the advantages and characteristics of the proposed model.

We compared our Mamba-based models with other remote sensing foundation models based on ViT-B, ViT-L, and Swin-B in terms of parameter size and throughput. As shown in Table I, the experiments were conducted using 1024-size images in the same environment. The RingMamba model, with VMamba as the backbone, has a parameter size of 88 million, which is comparable to Swin-B-based remote sensing models (e.g., RingMo [1] and Satlas [46]). However, our throughput is twice as high. Similarly, the RingMamba model with ASSCMamba as the backbone has a parameter size of 320 million, which is similar to ViT-L-based models like SatMAE [8], but achieves 2.6 times higher throughput. These results demonstrate that our Mamba-based framework introduces innovative adaptations tailored to remote sensing scenarios. By leveraging this novel framework, we significantly enhance throughput and computational efficiency while maintaining parameter sizes comparable to other SOTA models.

TABLE I

Throughput and Parameters Comparison of Common Remote Sensing Foundation Models Processing 1024-Size Images on an NVIDIA A100 (80 GB) GPU With a Batch Size of 32

Backbone	Model	Throughput ↑	Params
ViT-B	RVSA	23	89M
ViT-L	SatMAE, Scale-MAE	5	310M
Swin-B	RingMo, Satlas, GFM, CMID	13	87M
VMamba-B	RingMamba	25	88M
ASSCMamba	RingMamba	13	320M

TABLE II

Basic Information of Our Multimodal Pretraning Dataset

Mode	Channel	Paired	Number	Size
Optical	3	Yes	500,000	224×224
SAR	1	Yes	500,000	224×224
Optical	3	No	1,000,000	224×224
SAR	1	No	1,000,000	224×224

A. Pretraining Stage

1) Multimodal Pretraining Dataset: According to the imaging mechanism, we select representative types from two major categories of sensors: optical sensors and SAR sensors. Optical images can capture detailed color and texture information of ground targets, while SAR images can reveal structural information of ground areas. Similar to RingMo [1], the pretraining dataset constructed in this article is sourced from public datasets in the field of RS. The optical data mainly comes from Gaofen-1 and Gaofen-2 satellites, and the SAR data comes from the Gaofen-3 satellite, with resolutions ranging from 1 to 5 m (see Table II for details). To enhance the multisource fusion and generalization capabilities, the pretraining dataset is split into two segments: paired multisensor data (from the same area) and unpaired multisensor data (from different areas).

2) Experimental Settings: Based on the VMamba structure, we employ a multisensor self-supervised pretraining method. To evaluate the effectiveness of the proposed framework, we conducted comparative experiments involving various pretraining strategies and model modifications. Specifically, we compared the results of using natural scene pretraining parameters, pretraining on remote sensing data, incorporating LBP loss, and making structural adjustments to the model.

All experiments on the base model are implemented using NVIDIA A100 GPUs. The input size is 192. The mask ratio is 0.6. The mask patch size is set to 32. We use the AdamW optimizer with a momentum of 0.9 and a weight decay of 0.05 to optimize the loss. The optimizer betas are set to (0.9, 0.999). The base learning rate is set to 2e-4. We apply a warm-up period of 10 epochs before pretraining for a total of 200 epochs. In the experimental results, "RSSP" represent "remote sensing self-supervised pretraining without LBP loss." Bold text highlights the best result within each experimental group, while "*" indicates the top result across all methods.

3) Visualized Analysis: Fig. 6 illustrates some of the visual results from RingMamba during the pretraining phase. It is apparent that the reconstructed images exhibit some blurred regions when compared to the original images. This blurring occurs because the masked images have lost significant pixel information, which inherently complicates the reconstruction process. Despite this challenge, RingMamba effectively recovers the key semantic regions in both optical and SAR scenes, demonstrating that our model has successfully acquired general feature characterization capabilities through the

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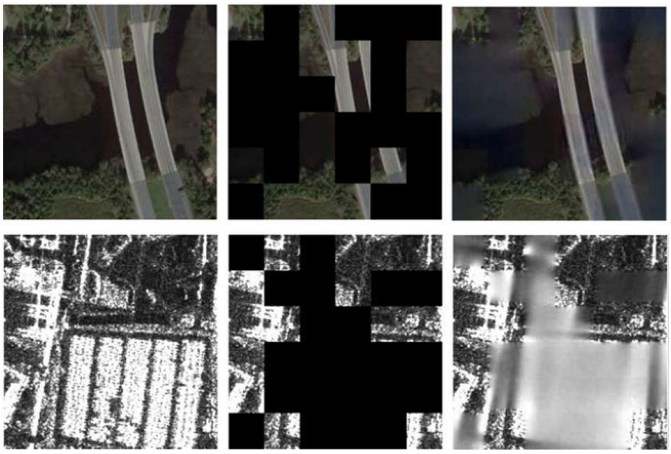


Fig. 6. Here are some visualized results from the pretraining stage. Each batch of examples displays images from left to right: the input image, the masked image, and the fusion image.

MM-MIM framework. This achievement is largely due to the integration of the multisensor loss constraint and the ASSC block, which effectively addresses the unique characteristics of remote sensing images. These visual results provide a clear demonstration of the effectiveness and robustness of our RingMamba model.

TABLE III

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE WHU-OPT-SAR DATASET FOR MULTIMODAL SEMANTIC SEGMENTATION. OA, MIOU, AND MF1 ARE USED AS EVALUATION METRICS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Backbone	OA(%) ↑	mIoU(%) ↑	mF1(%) ↑
PSCNN [47]	-	79.7	50.7	64.8
CFNet [48]	MobileNet-V2	79.9	52.7	66.2
MBFNet [5]	ResNet-50	81.5	51.1	64.9
V_FuseNet [6]	ResNet-50	81.6	48.2	61.0
MRSDC [7]	-	81.9	54.2	68.2
DDHRNet [49]	HRNet-48	82.0	52.3	66.6
FuseNet [50]	ResNet-50	82.2	54.2	63.7
MCANet [51]	ResNet-101	82.7	58.3	71.7
CMFNet [52]	VGG16	81.1	52.1	66.5
mmFormer [53]	-	83.8	55.1	67.0
TransFuser [54]	ResNet-50	83.9	61.8	74.9
TokenFusion [55]	MiT-b1	84.0	**62.3**	**75.0**
CEN [56]	ResNet-50	84.2	59.8	73.4
RingMamba	VMamba-B	83.4	53.3	67.2
RingMamba	ASSCMamba	**85.5**	59.3	73.2

B. Multimodal Segmentation

1) Dataset Introduction: To assess the model's performance and its ability to generalize in the task of multimodal segmentation, we have carried out a series of rigorous experiments utilizing the dataset: WHU-OPT-SAR.

1) *WHU-OPT-SAR [51]:* The dataset contains 100 pairs of optical/SAR images for land classification, making it larger than other datasets in the same category. The optical images include RGB and near-infrared (NIR) bands with a resolution of 5556×3704 pixels. The optical and SAR images in the dataset are from the GF-1 and GF-3 satellites, respectively. The dataset encompasses seven categories: farmland, road, urban, forest, rural, water, and others. The number of training, validation, and test sets is 17640, 5880, and 5880, respectively.

2) Experimental Settings: We replace only the encoder of UPerNet [57] with different pretrained models to perform the multimodal segmentation. During the training, we use the AdamW optimizer with a learning rate of 2e-4 and weight decay of 0.01. To ensure consistency with prior studies, we train the network for 160000 iterations using a batch size of 96, and we utilize overall accuracy (OA), the Kappa coefficient (Kappa), mean intersection over union (mIoU), and mean $F1$ score (mF1) as metrics to provide a more comprehensive assessment of performance. Experiments on this dataset are conducted on a single NVIDIA A100 GPU.

3) Result Analysis: Table III reports the quantitative experimental metrics. Although RingMamba still has a certain gap compared with the SOTA methods in terms of mIoU and mF1, it still comprehensively outperforms the mainstream methods in the table, such as MCANet and mmFormer. Besides, when only deploying the simple UperNet as the decoder, our model achieves the most advanced performance on the other two metrics, OA and Kappa, which fully demonstrates the superiority of RingMamba. Fig. 7 displays some visualized multimodal segmentation results. Obviously, our model can obtain more precise results in the details of region boundaries. Therefore, the quantitative and qualitative experimental results on the multimodal segmentation task fully demonstrate the advancement of the proposed MM-MIM framework and RingMamba.

C. Scene Classification

1) Dataset Introduction: Two widely used RS scene classification datasets are adopted to evaluate the performance.

1) *AID [64]:* The AID dataset comprises 10000 images, each measuring 600×600 pixels, with spatial resolutions between 0.5 and 8 m. It is divided into 30 categories, each containing about 220–400 images. In our experiments, we adhere to standard protocols by utilizing 20% and 50% for training, while the rest is reserved for testing.
2) *NWPU-RESISC45 [65]:* The dataset consists of 31500 images, each measuring 256×256 pixels. It is categorized into 45 classes, with each class containing 700 images. Following the protocols established in previous studies [1], [22], we allocate 10% and 20% for training, reserving the remaining 90% and 80% for testing, respectively.

2) Experimental Settings: All experiments were conducted utilizing an NVIDIA A100 GPU and a batch size of 64 [31]. The Adam optimizer was employed for model optimization

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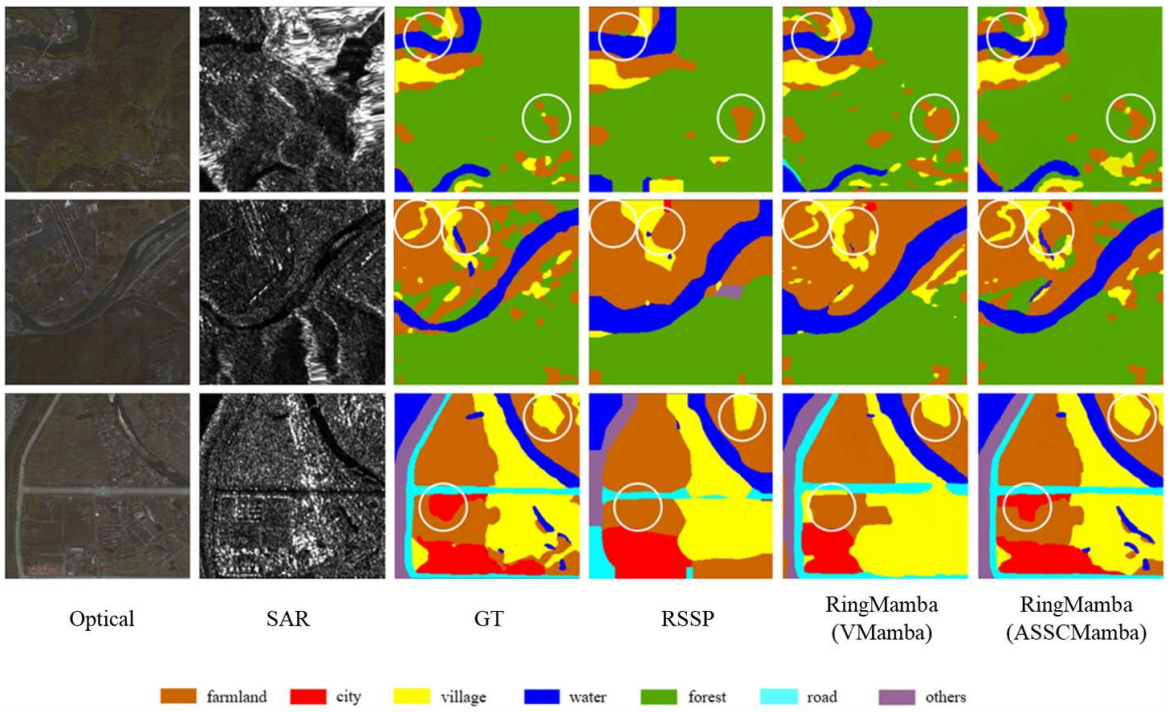


Fig. 7. Above are some visualized results for the multimodal segmentation task.

TABLE IV

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE AID AND NWPU-RESISC45 DATASET FOR SCENE CLASSIFICATION. OA IS USED AS EVALUATION METRICS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Model	Backbone	Image Size	Params	FLOPs ↓	AID TR=20%/50%	NWPU-RESISC45 TR=10%/20%
					OA ↑	OA ↑
CACo [58]	ResNet-18	224^2	11M	2G	90.88 / 95.05	88.28 / 91.94
GASSL [59]	ResNet-50	224^2	24M	4G	93.55 / 95.92	90.86 / 93.06
SeCo [60]	ResNet-50	224^2	24M	4G	93.47 / 95.99	89.64 / 92.91
TOV [15]	ResNet-50	224^2	24M	4G	95.16 / 97.09	90.97 / 93.79
SSL4EO [61]	ResNet-50	224^2	24M	4G	91.06 / 94.74	87.60 / 91.27
RVSA [62]	ViT-B + RVSA	224^2	86M	17G	**97.03** / **98.50**	93.93 / 95.69
SatMAE [8]	ViT-L	224^2	300M	62G	95.02 / 96.94	91.72 / 94.10
Scale-MAE [19]	ViT-L	224^2	300M	62G	96.44 / 97.58	92.63 / 95.04
RingMo [1]	Swin-B	224^2	87M	16G	96.90 / 98.34	**94.25** / 95.67
CMID [63]	Swin-B	224^2	87M	16G	96.11 / 97.79	94.05 / 95.53
SatLas [46]	Swin-B	224^2	87M	16G	94.96 / 97.38	92.16 / 94.70
GFM [18]	Swin-B	224^2	87M	16G	95.47 / 97.09	92.73 / 94.64
RingMamba	VMamba-B	224^2	88M	15G	94.24 / 96.94	90.22 / 92.63
RingMamba	ASSCMamba	224^2	320M	54G	96.54 / 97.82	93.70 / **95.71**

until convergence. We enhance our model by integrating a classification head specifically designed for the classification task, utilizing cross-entropy loss for optimization. OA is the primary metric in scene classification.

3) Result Analysis: Table IV reports the quantitative experimental metrics. Obviously, our RingMamba can effectively facing the different training-testing divisions and achieve the competitive performance. For the NWPU-RESISC dataset, although the performance of our model slightly lags behind the SOTA method at a 10% ratio, it still obtains the highest accuracy at a 20% ratio. Therefore, these quantitative metrics intuitively prove that our RingMamba is competitive with the existing mainstream vision foundation model, such as Swin transformer and ViT. We think this can be attributed to two aspects: 1) we introduced the MM-MIM pretraining framework to help our model fully perceive the robust prior knowledge and 2) we proposed the ASSC strategy to help our model precisely capture the characteristics of RS images.

D. Object Detection

1) Dataset Introduction: Experiments were conducted on the DIOR dataset and the SAR-AIRcraft-1.0 dataset, both of them are annotated with axially aligned bounding boxes.

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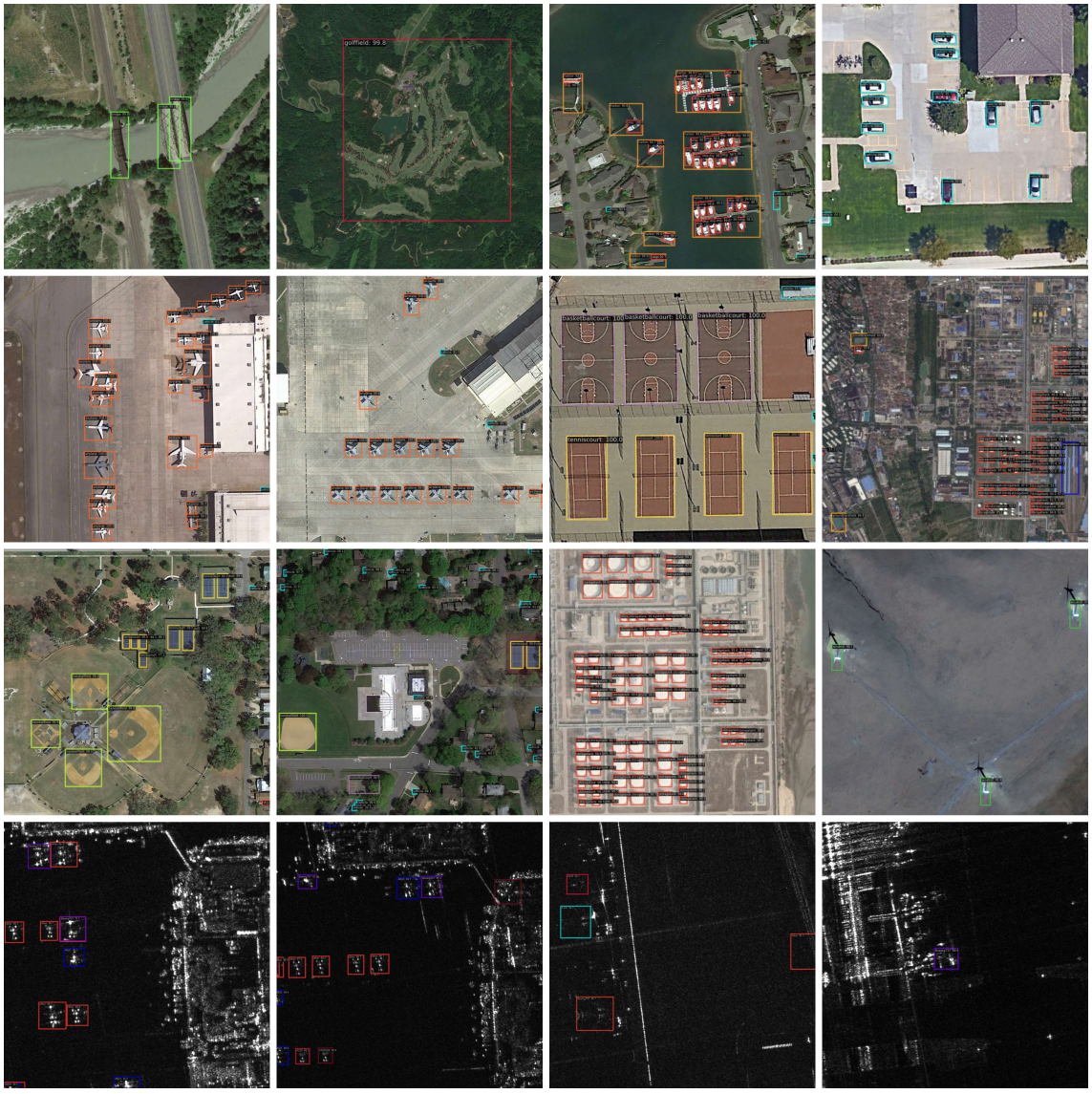


Fig. 8. Visualization results of the proposed model on the tasks of optical image target detection and SAR image target detection. It can be seen that the model has a good ability to locate and identify targets.

1) *DIOR [68]:* The dataset is designed for large-scale object detection in RS images. It includes 23463 images and 190288 instances, labeled with axially aligned bounding boxes across 20 categories.
2) *SAR-AIRcraft-1.0 [66]:* It is a high-resolution SAR dataset tailored for aircraft recognition and detection in complex environments. It includes 4368 images at a 1-m resolution, featuring annotations for 16463 aircraft instances across seven categories.

2) Experimental Settings: We replace only the encoder of Faster R-CNN [69] with different pretrained models to perform the object detection. During the training process, we used the AdamW optimizer with a weight decay of 0.01. We set the learning rate to 0.0001 and trained for 72 epochs. The LinearLR scheduler was applied in the first 500 iterations to warm up the model, starting with a factor of 0.001. We reduced the learning rate to 0.1 times its value after epochs 48 and 66. The height and width of all images were randomly scaled to a range of 800–1200 for multiscale training.

3) Result Analysis: Tables V and VI report the quantitative experimental metrics. For the DIOR dataset, compared with the existing advanced foundation model-based methods, such as SatMAE and RingMo, our model achieves the significant performance, which fully demonstrates the advancement of RingMamba. For the SAR-AIRcraft-1.0 dataset, although our RingMamba still has a certain gap compared with the advanced DiffDet4SAR, it still outperforms the mainstream methods in the table. In particular, different from DiffDet4SAR, whose performance is severely degraded under the certain parameter setting, our model can achieve the stable performance improvement by only replacing the backbone of Faster R-CNN, which also fully indicates the outstanding generalization ability of RingMamba. Compared with representative foundation models, RingMamba achieves noticeably better performance on the SAR-AIRcraft-1.0 dataset. This

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TABLE V
QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE SAR-AIRCRAFT-1.0 DATASET FOR OBJECT DETECTION. MAP50 IS USED AS EVALUATION METRICS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Backbone	mAP50(%) ↑
Specialized Model		
Faster R-CNN [66]	ResNet-50	76.1
Cascade R-CNN [66]	ResNet-50	75.7
Reppoints [66]	ResNet-50	72.6
SKG-Net [66]	ResNet-50	70.7
SA-Net [66]	ResNet-50	77.7
DiffDet4SAR (scale=1) [67]	ResNet-50	**88.4**
DiffDet4SAR (scale=0.1) [67]	ResNet-50	73.1
Foundation Model		
CACo [58]	ResNet-18	16.8
GASSL [59]	ResNet-50	54.9
SSL4EO [61]	ResNet-50	49.2
SatLas [46]	Swin-B	56.3
CROMA [68]	ViT-B	68.4
RingMamba	VMamba-B	78.3
RingMamba	ASSCMamba	82.0

further highlights the advantage of our multimodal pretraining framework, which better captures SAR-specific features and improves cross-modal generalization. Fig. 8 further reports some visualization results. Obviously, our model can accurately detect the interest objects with different situations and scales. Therefore, the quantitative and qualitative experimental results on the object detection task once again fully proves the advancement of our framework.

E. Semantic Segmentation

1) Dataset Introduction: To assess the generalization of the model in semantic segmentation tasks, we performed experiments on three datasets: ISPRS Potsdam, iSAID, and Air-PolSAR-Seg.

1) *ISPRS Potsdam Dataset [86]:* The dataset includes 38 high-resolution aerial images, each measuring 6000 × 6000 pixels with a resolution of 5 cm. It is annotated with six common feature classes: buildings, trees, cars, impervious surfaces, low vegetation, and clutter.
2) *iSAID Dataset [87]:* The dataset comprises 2806 high-resolution images containing 655451 object instances across 15 common aerial imagery classes. As the test set annotations are not publicly available, we use the validation set for testing purposes in our experiments.
3) *Air-PolSAR-Seg Dataset [88]:* The dataset is a comprehensive, large-scale dataset designed for semantic segmentation in fully polarimetric synthetic aperture radar (PolSAR) images of complex scenes. It includes six categories: housing, industrial, water, land use, natural, and other areas. These images were captured by the GF-3 satellite operating in quad-polarized strip I (QPSI) mode. The dataset comprises 500 images, each with a resolution of 8 m and dimensions of 512 × 512 pixels. For training, we selected 1400 image patches, leaving the remaining 600 patches for the test set.

2) Implementation Details: We utilized two NVIDIA A40 GPUs for the AIR-PolSAR-Seg and Potsdam datasets, and one NVIDIA A100 GPU for the iSAID dataset. In our experiments, the images from the iSAID dataset were cropped to 896×896 pixels, while the images from the Potsdam and AIR-PolSAR-Seg datasets were cropped to 512×512 pixels. The batch size was adjusted according to the dataset: two for the Potsdam dataset and four for both the AIR-PolSAR-Seg and iSAID datasets.

For optimization, we used the AdamW optimizer across all three datasets and implemented UPerNet [57] with cross-entropy loss for the output head. The initial learning rate and weight decay were set as follows: for the iSAID dataset, the initial learning rate was $1e^{-4}$ with a weight decay of 0.05; for the AIR-PolSAR-Seg and Potsdam datasets, the initial learning rate was $6e^{-5}$ with a weight decay of 0.01. Each model was trained for 160k iterations, starting with 1500 rounds of linear warm-up, followed by a polynomial decay strategy.

3) Result Analysis: Tables VII and VIII report the quantitative experimental metrics. For the iSAID dataset, although our RingMamba does not obtain the highest score, it still outperforms the mainstream segmentation models when only deploying the simple UperNet as the decoder, For ISPRS Potsdam and Air-PolSAR-Seg datasets, compared with the existing advanced methods based on other foundation models, our model still achieves the most advanced performance. Particularly, the performance improvement of our RingMamba on the Air-PolSAR-Seg dataset is significant. Fig. 9 shows some visualization results. Obviously, our proposed method has higher accuracy in small target segmentation, and also has significant advantages in the stability and efficiency of edge segmentation. Therefore, the quantitative and qualitative experimental results on the semantic segmentation task again effectively reflects the advancement of RingMamba.

F. Change Detection

1) Dataset Introduction: The LEVIR-CD dataset was used in this article. It consists of 637 pairs of RS images, each with a resolution of 1024 × 1024 pixels. For a fair comparison, we adopted a standard data processing approach [89], cropping the images into nonoverlapping patches of 256 × 256 pixels.

2) Implementation Details: We replace only the encoder of BIT [90] with different pretrained models to perform the change detection. Optimization is carried out using AdamW, with a weight decay of 0.01 and beta values set to (0.9, 0.999). The initial learning rate is 0.0001, which decays to 0 over 200 epochs. We use $F1$ score, IoU, and OA as the final evaluation metrics.

3) Result Analysis: Table IX and Fig. 10 display the results of our experiments. Specifically, although the recent Mamba-based methods, such as ChangeMamba and CDMamba, have

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TABLE VI

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE DIOR DATASET FOR OBJECT DETECTION. MAP50 IS USED AS EVALUATION METRICS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Backbone	Image Size	Params	FLOPs ↓	mAP50(%) ↑
CACo [58]	ResNet-18	800^2	28M	101G	66.9
GASSL [59]	ResNet-50	800^2	41M	134G	67.4
TOV [15]	ResNet-50	800^2	41M	134G	70.2
SSL4EO [61]	ResNet-50	800^2	41M	134G	64.8
RVSA [62]	ViT-B + RVSA	800^2	113M	378G	73.22
SatMAE [8]	ViT-L	800^2	324M	1094G	70.9
Scale-MAE [19]	ViT-L	800^2	324M	1094G	73.8
RingMo [1]	Swin-B	800^2	105M	322G	75.9
CMID [63]	Swin-B	800^2	105M	322G	75.1
SatLas [46]	Swin-B	800^2	105M	322G	74.1
GFM [18]	Swin-B	800^2	105M	322G	72.8
RingMamba	VMamba-B	800^2	107M	202G	77.60
RingMamba	ASSCMamba	800^2	336M	738G	**78.30**

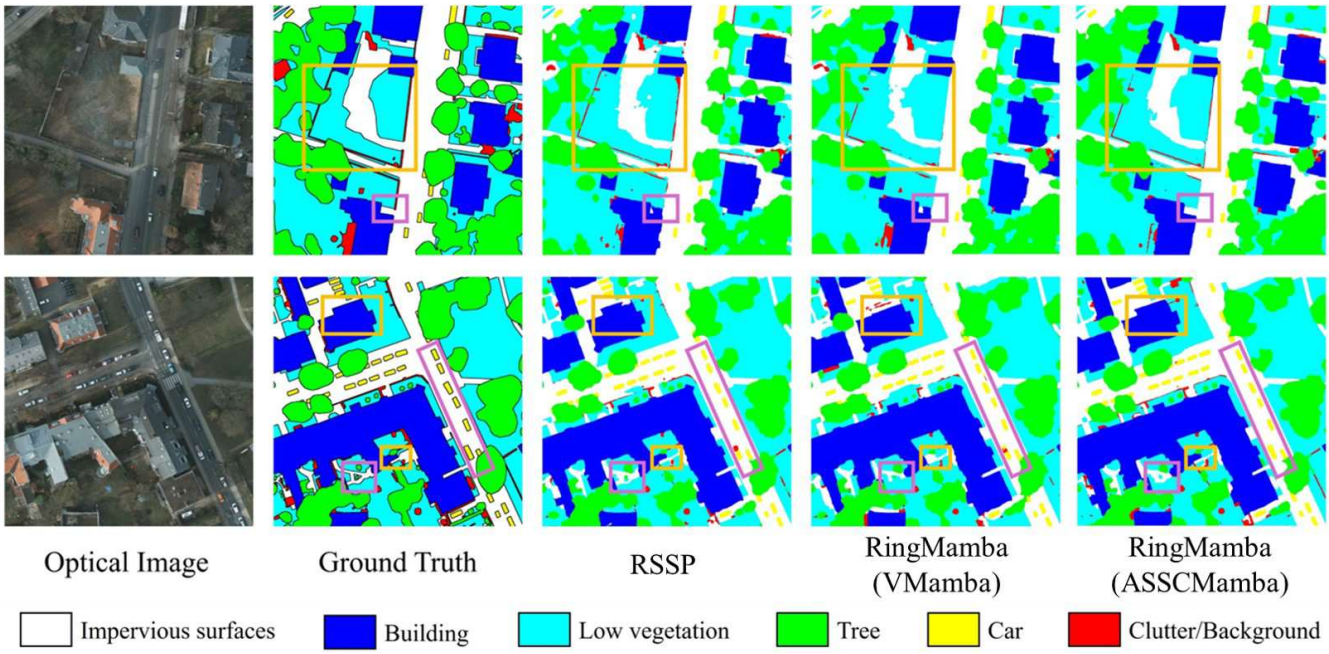


Fig. 9. Experimental results are visualized on the ISPRS Potsdam dataset. It can be seen that our method has a better classification effect on small-sized objects and edge regions.

achieved preliminary success, they still lag behind some advanced change detection methods like MDIPNet. Our proposed MM-MIM and ASSCMamba significantly boost BIT's performance, showcasing the effectiveness of our framework. Furthermore, the visual examples in Fig. 10 demonstrate that our model provides more accurate predictions on the detail boundaries of changed regions. Therefore, these results further underscore the superiority and versatility of RingMamba.

G. Ablation Analysis

1) Ablation on MM-MIM Framework: In this study, we propose the MM-MIM framework to enable remote sensing pretraining under multimodal conditions, with a primary focus on enhancing the loss function to facilitate cross-modal information interaction. To validate the effectiveness of the proposed framework, we incorporate ASSCMamba both before and after loss-based modification, allowing a comprehensive assessment of its impact. Specifically, we conduct evaluations on four mainstream downstream tasks: multimodal segmentation, object detection, change detection, and semantic segmentation to ensure a thorough exploration of the framework's capabilities. As demonstrated by the quantitative results in Table X, our framework consistently improves performance in all downstream tasks, providing strong evidence of the effectiveness of MM-MIM.

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TABLE VII

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON iSAID AND ISPRS POTSDAM DATASET FOR SEMANTIC SEGMENTATION. mIoU AND mF1 ARE USED AS VALUATION METRICS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Backbone	Image Size	Params	FLOPs ↓	iSAID	Potsdam
					mIoU(%) ↑	mF1(%) ↑
CACo [58]	ResNet-18	896^2	41M	671G	64.32	91.35
GASSL [59]	ResNet-50	896^2	64M	722G	65.95	91.27
SeCo [60]	ResNet-50	896^2	64M	722G	57.20	89.03
TOV [15]	ResNet-50	896^2	64M	722G	66.24	92.03
SAMRS [71]	ResNet-50	896^2	64M	722G	66.26	91.43
SSL4EO [61]	ResNet-50	896^2	64M	722G	64.01	91.54
RVSA [62]	ViT-B + RVSA	896^2	128M	1043G	64.49	91.15
SatMAE [8]	ViT-L	896^2	341M	1536G	62.97	90.63
Scale-MAE [19]	ViT-L	896^2	341M	1536G	65.77	91.54
RingMo [1]	Swin-B	896^2	121M	968G	67.20	91.27
GFM [18]	Swin-B	896^2	121M	968G	66.62	91.85
CMID [63]	Swin-B	896^2	121M	968G	66.21	91.86
SatLas [46]	Swin-B	896^2	121M	968G	**68.71**	91.28
BFM [72]	ViT-G12	896^2	-	-	-	92.12
RingMamba	VMamba-B	896^2	121M	889G	64.99	92.27
RingMamba	ASSCMamba	896^2	350M	1511G	66.42	**92.70**

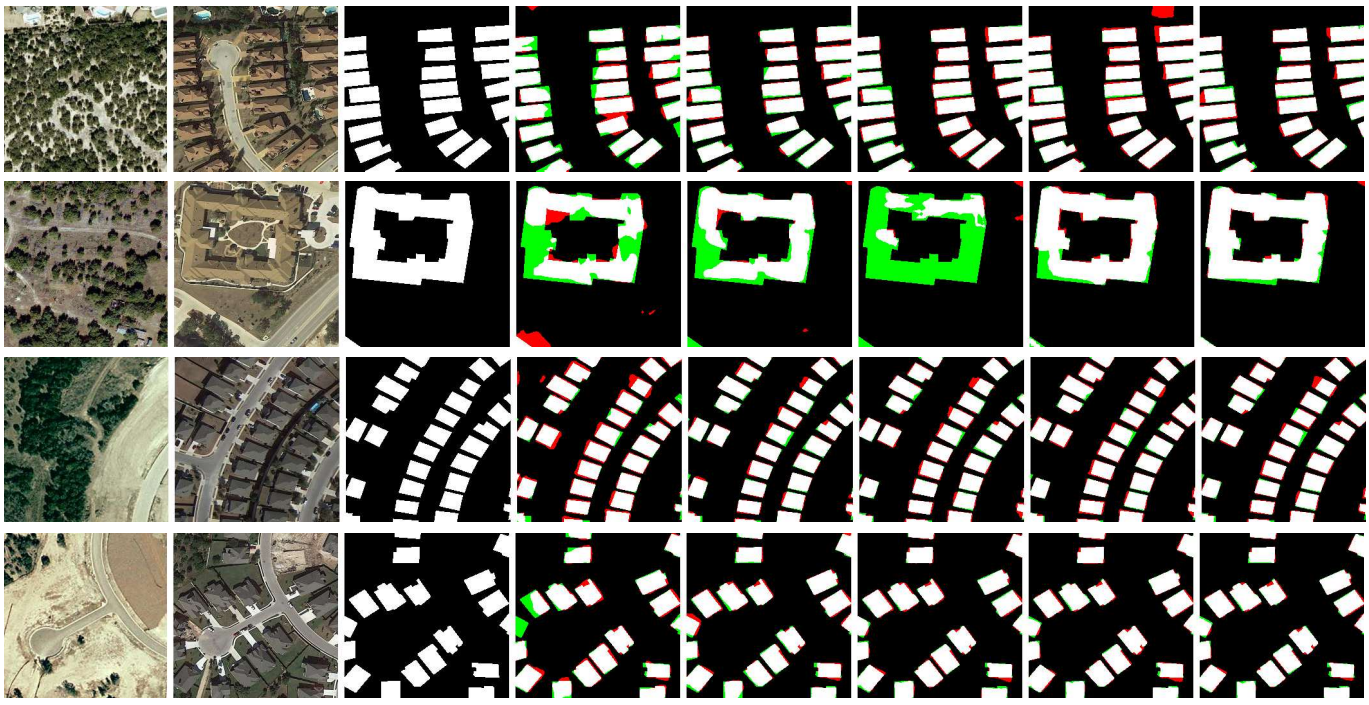


Fig. 10. Below are some visualized results for the change detection task. For each batch of examples, the images are arranged from left to right as follows: pretime image, posttime image, ground truth, BIT prediction, BIT-RSSP prediction, BIT-RingMamba prediction, and BIT-RingMamba-ASSCMamba prediction. The regions are color-coded with black indicating true negative, white indicating true positive, green indicating false negative, and red indicating false positive.

2) Ablation on ASSCMamba: In this study, we propose ASSCMamba, which incorporates ASSC modules to better accommodate the unique characteristics of RS images. To verify their effectiveness, we performed ablation studies by individually removing these two modules. Following the same evaluation protocol, we validate the model in four main downstream tasks: multimodal segmentation, object detection, change detection, and semantic segmentation, ensuring a comprehensive analysis. As shown in Table XI, the proposed modules significantly improve the performance of VMamba in

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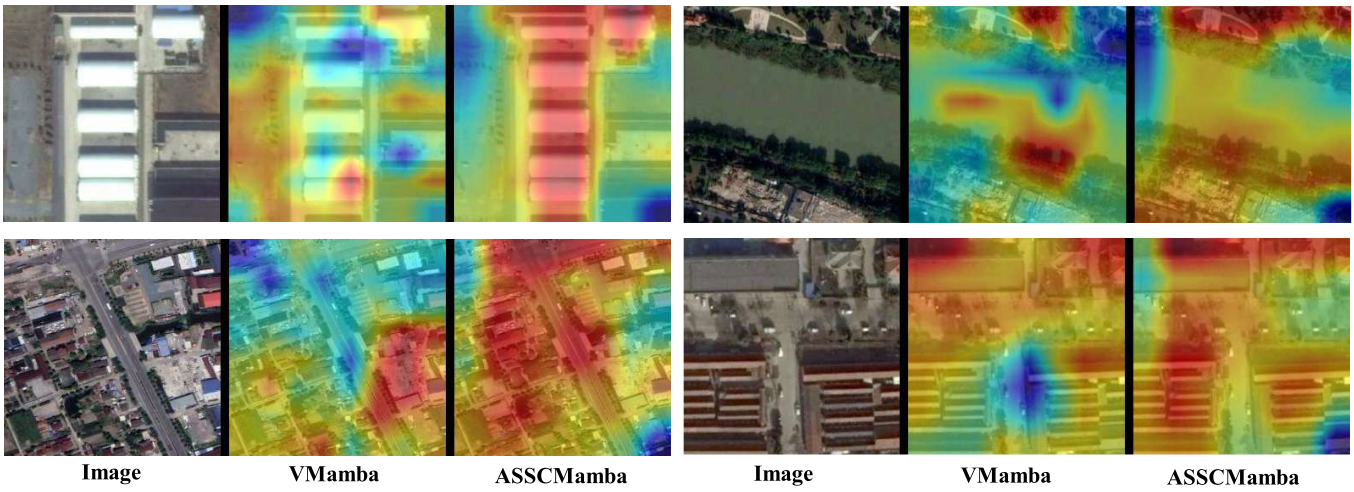


Fig. 11. Feature visualization comparison between Vmamba and ASSCMamba. The feature response is displayed as a heat map. The higher the heat value, the higher the feature response. Obviously, our proposed ASSC module can perceive the global ground information more precisely.

TABLE VIII

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE AIR-POLSAR-SEG DATASET FOR SEMANTIC SEGMENTATION. MIOU AND OA ARE USED AS EVALUATION METRICS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Backbone	mIoU(%) ↑	OA(%) ↑
Specialized Model			
FCN [73]	VGG-16	48.35	76.28
ANN [74]	ResNet-50	52.58	77.46
DeepLab V3+ [75]	ResNet-50	48.21	76.81
EmaNet [76]	ResNet-50	47.56	76.36
U-Net [77]	—	52.12	77.29
CCNet [78]	ResNet-50	46.46	75.53
DANet [79]	ResNet-50	51.93	76.91
EncNet [80]	ResNet-50	47.75	75.67
PSANet [81]	ResNet-50	47.14	76.21
GCNet [82]	ResNet-50	47.56	76.75
PSPNet [83]	ResNet-50	51.48	76.55
NonLocal [84]	ResNet-50	44.23	76.05
FUSAR-Map [85]	Xception-65	55.45	74.42
MASA-SegNet [86]	MAXIM	54.25	77.60
Foundation Model			
CACo [58]	ResNet-18	41.63	74.25
GASSL [59]	ResNet-50	42.17	73.38
SSL4EO [61]	ResNet-50	42.03	74.27
SatLas [46]	Swin-B	47.51	74.98
CROMA [68]	ViT-B	51.32	76.56
RingMamba	VMamba-B	54.58	77.31
RingMamba	ASSCMamba	**57.42**	**81.19**

TABLE IX

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE LEVIR-CD DATASET FOR CHANGE DETECTION. $F1$ IS USED AS EVALUATION METRICS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Backbone	Params	FLOPs ↓	F1(%) ↑
Foundation Model				
CACo [58]	ResNet-18	12M	11G	81.04
GASSL [59]	ResNet-50	27M	25G	78.19
SSL4EO [61]	ResNet-50	27M	25G	89.05
SatMAE [8]	ViT-L	304M	162G	87.65
SatLas [46]	Swin-B	88M	45G	90.62
Specialized Model				
AMTNet-50 [92]	ResNet-50	25M	22G	90.76
ChangeFormer [26]	MiT-B2	41M	811G	90.40
ChangeCLIP [93]	ViT-B	171M	79G	91.30
MDIPNet [94]	Swin-T	40M	28G	91.49
BIT [95]	ResNet-18	12M	106G	90.14
RSMamba [37]	OSSMamba	28M	16G	89.77
ChangeMamba [38]	MambaBCD	85M	179G	90.16
CDMamba [96]	SRCMamba	9M	54G	90.75
Ours				
RingMamba	VMamba-B	113M	55G	91.52
RingMamba	ASSCMamba	372M	172G	**91.65**

the four tasks. For example, ASSCMamba improves the mF1 score in multimodal segmentation by 6.0 points and increases the mAP50 in object detection by 3.7 points compared to the original VMamba. Fig. 11 further displays some feature visualization results. Obviously, compared with the original VMamba, our ASSCMamba can precisely perceive the ground information from global perspectives, which intuitively shows the effectiveness of the proposed ASSC module. Therefore, these ablation results strongly confirm the rationality of this study.

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TABLE X
ABLATION STUDY ON THE NECESSITY OF MM-MIM

Ablation operation	Multimodal Segmentation			Object Detection	Change Detection		Semantic Segmentation
	OA(%)	mIoU(%)	mF1(%)	MAP50(%)	F1(%)	IoU(%)	mF1(%)
RSSP-ASSCMamba	84.40	58.90	72.80	81.50	91.64	84.57	92.49
RingMamba-ASSCMamba	**85.50**	**59.30**	**73.20**	**82.00**	**91.65**	**84.58**	**92.70**

TABLE XI
ABLATION STUDY ON THE NECESSITY OF ASSCMAMBA

Ablation operation	Multimodal Segmentation			Object Detection	Change Detection		Semantic Segmentation
	OA(%)	mIoU(%)	mF1(%)	MAP50(%)	F1(%)	IoU(%)	mF1(%)
RingMamba-VMamba	83.40	53.30	67.20	78.30	91.52	84.38	92.27
RingMamba-ASMamba	84.00	54.30	68.30	81.10	91.56	84.44	92.42
RingMamba-ASSCMamba	**85.50**	**59.30**	**73.20**	**82.00**	**91.65**	**84.58**	**92.70**

V. CONCLUSION

In this study, we have presented RingMamba, an innovative multisensor remote sensing foundation model that excels in both performance and efficiency. The multisensor self-supervised pretraining framework we introduced effectively combines generative and contrastive learning strategies under texture feature constraints, enabling robust unsupervised training on large-scale multisensor datasets. Our extensive experiments on nine public datasets, covering tasks such as multimodal semantic segmentation, scene classification, object detection, and change detection, demonstrate the superior performance of RingMamba. The model consistently achieves competitive results across these varied applications. In addition, our evaluations of computational efficiency highlight the model's high throughput and low FLOPs, particularly under high-resolution inputs, which underscores its suitability for practical remote sensing applications.

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